scientific reports

OPEN

Analysis of the impact mechanisms and driving factors of urban spatial morphology on urban heat islands

Caiyi Huang[1], Ke Liu[1,2]✉, Tianyi Ma[3], Hang Xue[1], Pengfei Wang[1] & Lin Li[1]

The intensity of urban heat islands (UHI) is closely related to urban underlying surface characteristics. This study investigates the effects of underlying surfaces on land surface temperature (LST) in the highly urbanized central districts and rapidly developing peri-urban areas of Tianjin, China. Ten driving factors linked to urban spatial morphology, land surface, and socioeconomic conditions were selected from 2 and 3D perspectives. LST was derived using Landsat 8 and MODIS data from August 28, 2020. XGBoost and SHAP were applied to analyze the contributions of individual and interacting factors on LST. The results showed that: (1) In the central districts, the contributions of NDVI (0.89), BH (0.30), and Albedo (0.25) were the highest. In the peri-urban areas, the contributions of NDVI (0.90), Albedo (0.60), and NLI (0.42) were the highest. NDVI showed a significant negative correlation with LST in both areas, with a more pronounced cooling effect compared to other factors. (2) In the central districts, the combination of BH and BD had the greatest impact on LST, with cooling effects observed when BH > 20m and BD decreased. In the peri-urban areas, the combination of NDVI and PIS was most influential, with higher PIS increasing LST when NDVI < 0.5. (3) Urban spatial parameters had a weaker impact on LST in the peri-urban areas compared to the central districts, where socioeconomic parameters played a stronger role. Overall, 2D perspectives had a greater impact on LST than 3D perspectives. The findings highlight the need for tailored regulatory measures based on urban development stages.

Keywords Drivers, LST, SHAP, UHI, Urban spatial form, XGBoost

Abbreviations

BD	Building density
BH	Building height
BHD	Building height deviation
LST	Land surface temperature
NDVI	Normalized difference vegetation index
NDWI	Normalized difference water index
NLI	Night light index
PGS	Percentage of green space
PIS	Percentage of impervious surface
SHAP	SHapley additive explanations
SVF	Sky view factor
UHI	Urban heat island
XGBoost	Extreme gradient boosting

The urban heat island (UHI) effect refers to the phenomenon where the temperature of an urban core area rises significantly compared to its surrounding environment, due primarily to increased human construction activities and changes in land surface cover[1]. The formation and intensification of the UHI effect are driven mainly by changes in the spatial morphology of the urban subsurface, specifically in the layout and organisational characteristics of spatial elements such as buildings and green spaces[2,3]. Those changes alter the permeability and evapotranspiration properties of the land cover, subsequently affecting the regulation of surface temperature and humidity[4]. Therefore, exploring the effect of urban spatial morphology on UHIs and its driving mechanisms

[1]North China Institute of Aerospace Engineering, Langfang 065000, China. [2]Aerospace Remote Sensing Information Processing and Application Collaborative Innovation Center of Hebei Province, Langfang 065000, China. [3]Xiong'an Aerospace Information Research Institute, Xiongan 071000, China. ✉email: liuke1176@163.com

can provide scientific insights for mitigating the UHI effect and offer practical strategies for urban planning and sustainable development.

Although land surface temperature (LST) might not directly affect human thermal comfort as perceptibly as air temperature, it plays a vital role in regulating the urban microclimate and is closely correlated with air temperature[5,6]. Additionally, using remote sensing imagery, LST can be easily and continuously determined over large areas. That makes it a key indicator for studying the UHI effect and a critical factor in urban planning and environmental management[7].

Land cover and land use types are considered important factors influencing LST[8]. Different land use types, such as cropland, grassland, and built-up areas, have significant functional and spatial differences[9], directly affecting how heat is absorbed and emitted from the surface, leading to varying impacts on LST[10]. Land surface temperature is negatively correlated with green space composition[11] and positively correlated with impervious surfaces[12]. Studies have shown that adjusting the proportions of urban green spaces, water bodies[13], and impervious surfaces[14] is critical for urban planning and mitigating the UHI effect. A study by Zhou et al.[15] on the driving factors and spatial heterogeneity of UHIs in China found that daytime UHI intensity is influenced primarily by the vegetation difference between urban and rural areas, whereas at night, socioeconomic factors have a more significant effect. Therefore, in exploring the mechanisms linking land cover and land use types with LST, incorporating socioeconomic factors could provide a more comprehensive understanding of LST variation.

Recent studies have increasingly shifted from the 2D land cover perspective to the 3D building morphology perspective. Oke[1] pointed out that the roughness of urban subsurfaces further affects the local thermal environment by altering airflow and radiation transmission. Some scholars[16] have conducted quantitative evaluations of the UHI effect using 3D urban characteristics, such as building height (BH), building density (BD), floor area ratio, and the sky view factor (SVF), as driving factors influencing surface temperature.

However, discussions regarding the effect of 2D land cover parameters versus 3D building morphology parameters on LST remain contentious. Some studies[17] have concluded that 3D parameters have a significantly greater influence on LST than commonly used 2D parameters. In contrast, research on the seasonal differences in the influence of 2D/3D parameters on LST[18,19] has found that in summer, LST variations depend more on the influence of 2D parameters. It is evident that current research on factors driving LST must still consider the combined influence of both 2D and 3D urban built environments and their spatial interactions. Moreover, existing research has focused mainly on the spatiotemporal variations and driving mechanisms of the heat island effect in major cities or large urban agglomerations[20,22], with limited research on the effect of spatial morphology on the UHI effect in areas with various development orientations within cities.

In summary, considering that UHI is influenced by a combination of multidimensional parameters such as urban spatial morphology, land cover, and socioeconomic factors, this study selected 10 driving factors: BH, BD, building height deviation (BHD), the SVF, the normalised difference vegetation index (NDVI), the normalised difference water index (NDWI), the percentage of green space (PGS), the percentage of impervious surface (PIS), albedo, and the night light index (NLI). Based on multisource spatial data, the study used the eXtreme Gradient Boosting (XGBoost) model and the SHapley Additive exPlanations (SHAP) method to do a comparative analysis of the mechanisms driving UHIs in the central districts and peri-urban areas of Tianjin, China. The aim was to explore the internal relations between 2 and 3D urban built environments and between different development-oriented regions within the city and their relative importance to LST. The findings are intended to provide theoretical support and practical references for optimising urban spatial morphology and mitigating urban heat environments in the context of urban planning and development.

Material and methods

This study consists of four steps (see Fig. 1): (1) Collect and calculate 10 driving factors affecting urban heat islands, select Landsat 8 and MODIS data, derive LST using the windowing algorithm, and divide both factors and LST into 30 m × 30 m grids; (2) Define the urban development of Tianjin's central and peri-urban districts, distinguish their spatial characteristics and development stages, and establish a spatial reference for analysis; (3) Construct a dataset with the 10 factors as inputs and LST as the target variable, train the dataset using XGBoost, and evaluate the model performance; (4) Apply SHAP for interpretability, analyze the contributions of driving factors to LST from 3D (Urban spatial morphology) and 2D (land surface and socioeconomics) perspectives, and further explore the interactions between driving factors through SHAP interaction values to reveal the key drivers and complex patterns of the urban heat island effect.

Study area and data

Overview of the study area

Tianjin is located between 38°34' to 40°15' N latitude and 116°43' to 118°04' E longitude, governing 16 districts. Geographically, it can be divided into the central districts, the 4 peri-urban districts, the Binhai New Area, and the 5 outer suburban districts. As a key city in northern China, Tianjin's central districts (including Hongqiao, Hebei, Nankai, Heping, Hedong, and Hexi districts) are highly urbanised regions, whereas the 4 peri-urban districts of Beichen, Dongli, Xiqing, and Jinnan are rapidly urbanising areas at the urban–rural interface. This study selected Tianjin's central districts and the 4 peri-urban districts as the research area, which covered approximately 2082 km^2 (see Fig. 2).

The central district comprises densely populated, built-up old districts with a high concentration of buildings. As shown in the spatial distribution maps of driving factors in Fig. 3b, h, i, k, the area has a large proportion of impervious surfaces and relatively single land use types whose buildings serve primarily residential and commercial purposes. The area also lacks green space. The 4 peri-urban districts each have distinct development focuses: the Beichen District is a base for advanced manufacturing, technological innovation, and modern commercial logistics; the Xiqing District is a demonstration area for ecological civilisation and a

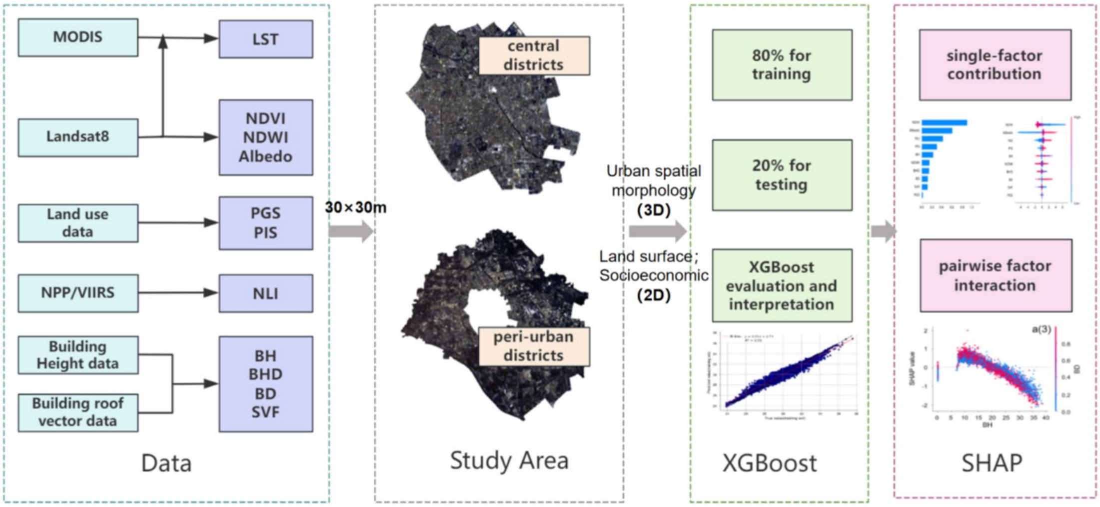


Fig. 1. Flow chart of the study. The map in this figure was generated using ArcGIS version 10.8 (ESRI, 2020). URL: https://www.esri.com/en-us/arcgis.

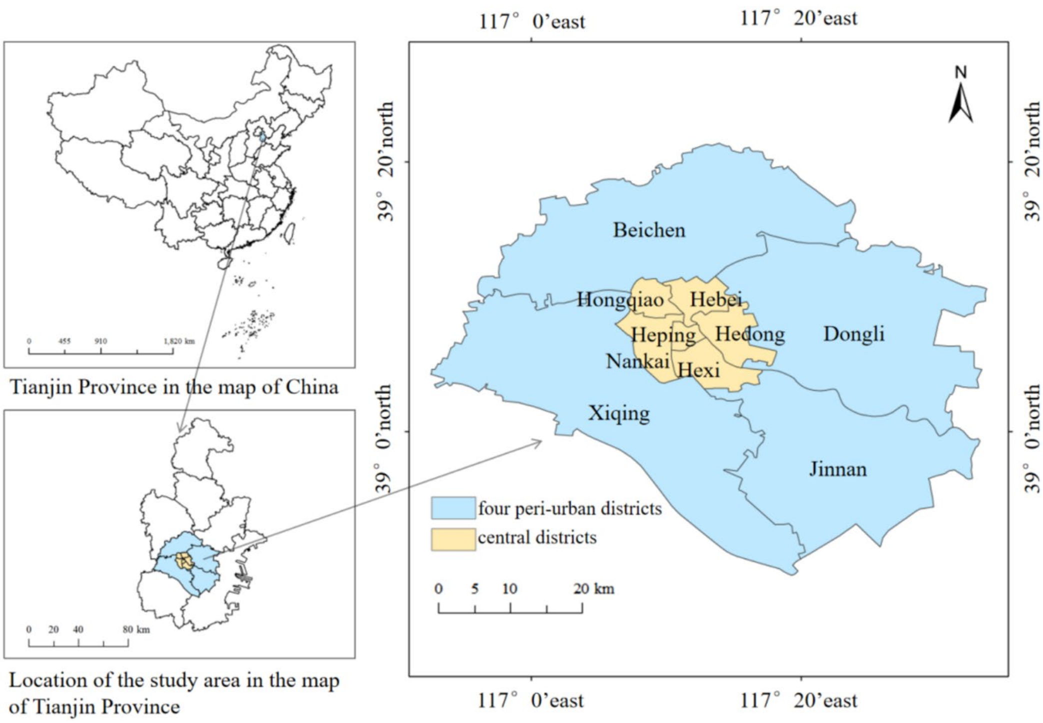
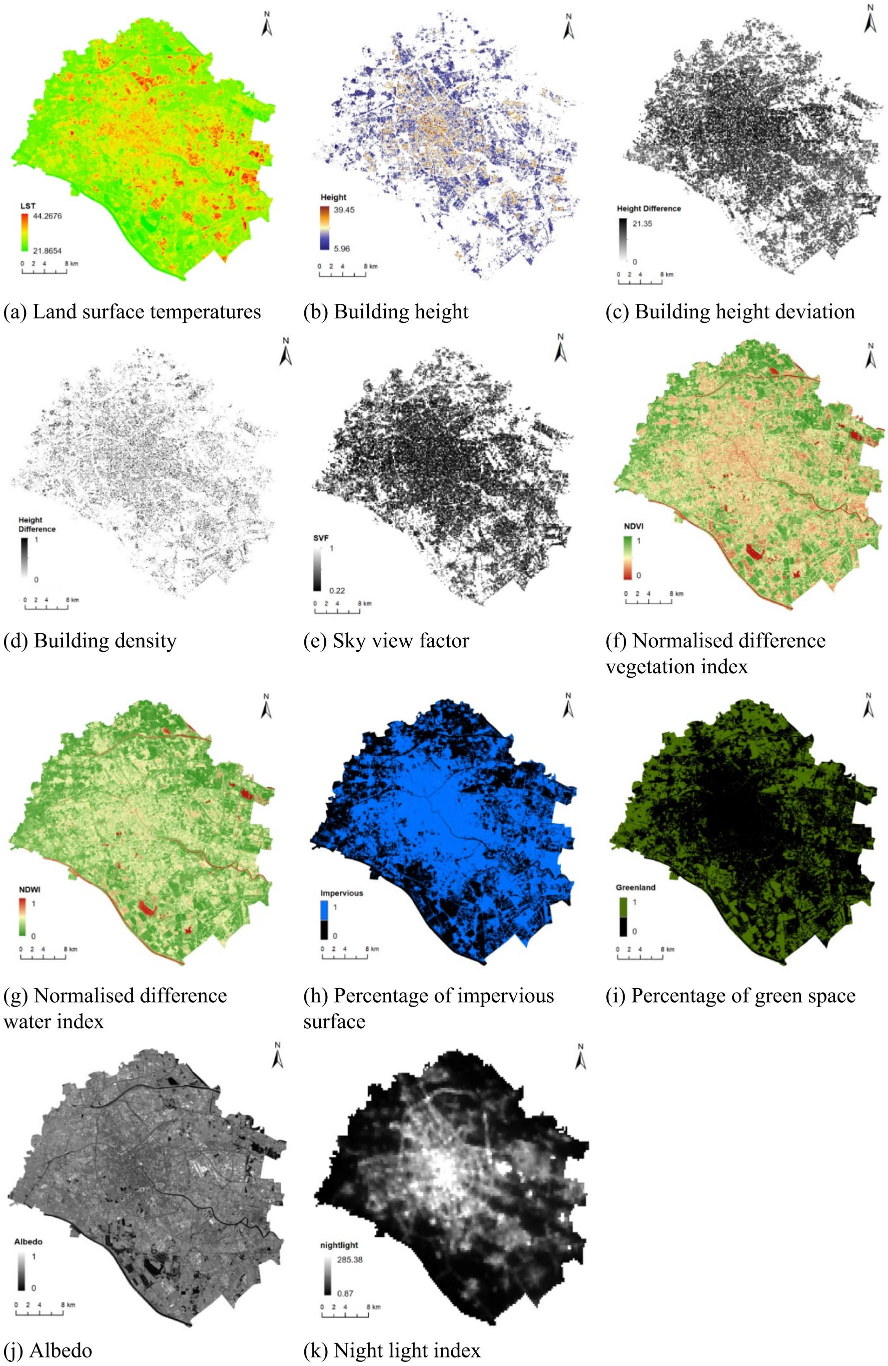


Fig. 2. Study area (The base map of China is sourced from the Tianditu platform (https://cloudcenter.tianditu.gov.cn/administrativeDivision/), with the approval number: GS (2024) 0650.)

base for innovation transformation within the Beijing–Tianjin–Hebei region, integrating industries such as the Internet, automobiles, new-energy vehicles, and high-tech sectors; the Dongli District is developing a modern comprehensive service area centred on its airport. It also serves as a hub for high-end manufacturing innovation and an international cultural exchange centre, and is a smart ecological city; and the Jinnan District serves as a national demonstration base for a higher education–industry collaboration, a hub for emerging industries, and is a new ecologically liveable district with integrated urban and industrial development. The varied development priorities of the 4 peri-urban districts, along with the heterogeneity of their economic development, provide valuable insights into the UHI effect that emerges under different geographic characteristics and levels of economic development within a city. This offers new perspectives and data to better understand the UHI effect that emerges during urban development.



(a) Land surface temperatures (b) Building height (c) Building height deviation (d) Building density (e) Sky view factor (f) Normalised difference vegetation index (g) Normalised difference water index (h) Percentage of impervious surface (i) Percentage of green space (j) Albedo (k) Night light index

Fig. 3. Spatial distribution of surface temperature and influencing factors (Generated using ArcGIS. 10.8)

Data availability

The data used in this study is shown in Table 1.

To ensure consistency in the spatial resolution of different data sources, this study standardized the datasets with varying resolutions. All data resampling processes were performed using the Nearest Neighbor method (NEAREST) in ArcGIS software, and all data were transformed into a unified coordinate system (WGS_1984) with a standardized spatial resolution of 30×30m for subsequent analysis and comparison.

Land surface temperature retrieval

The algorithms used to retrieve LST based on thermal infrared remote sensing can be divided into 3 general categories: an atmospheric correction method, a split-window algorithm, and a single-channel algorithm. Both the single-channel algorithm and the atmospheric correction method require real-time atmospheric profile data. However, those data are difficult to obtain, and as of 2024, NASA's official website no longer provides atmospheric water vapour content data. In this study, the split-window algorithm was used to retrieve LSTs. The

Data	Time	Spatial Resolution	Data Sources	Description
Landsat 8	2020.8.28,10:47 AM (local time)	30 m (multispect-ral); 100 m (thermal infrared)	USGS (https://earthexplorer.usgs.gov/) Scene ID: LC81220332020241LGN00	Preprocessing includes thermal infrared radiometric calibration,multispectral radiometric calibrat-ion, and atmospheric correctio-n;Thermal infrared band resa-mpled to match multispectral re-solution
MODIS MOD1B	2020.8.28,10:30 AM (local time)	1km	https://search.earthdata.nasa.gov/search scene ID: MOD021KM.A2020241.0215.061	Select the reflectance data from visible to shortwave infrared b-ands (bands 1–19, 26) , and p-erform geometric correction using MCTK (modis_toolkit). B-ilinear interpolation is then app-lied to resample the data to a 30 m resolution
MODIS LST	2020.8.28,10:30 AM (local time)	1km	https://search.earthdata.nasa.gov/search scene ID: MYD11A2.A2020241.h27v05.061.2021014172341	The MODIS LST data were re-projected and resampled to mat-ch the spatial resolution of the LST derived from Landsat 8, thereby validating the surface te-mperature retrieval results
Nighttime light data (NPP/VIIRS)	2020.8	500m	EOG(https://eogdata.mines.edu/products/vnl/)	Monthly nighttime light data fo-r August 2020 was selected to represent socioeconomic indica-tors, reflecting regional populat-ion distribution, economic deve-lopment levels, and energy con-sumption
Land use data	2020	30m	the Wuhan University Yang Jie team's 30 m annual land cover dataset (2020)[23] (https://zenodo.org/records/12779975)	It covers 9 major land cover ty-pes, including cropland, forest, shrub, grassland, water, snow/i-ce, barren, impervious, and we-tland,with an overall classificati-on accuracy of 80%;This study uses the dataset to extract info-rmation on green spaces (inclu-ding cropland, forest, and gras-sland) and impervious surfaces
Building roof vector data	2020	–	The vectorised rooftop area data for 90 cities in China (2020)[24] from the Chen Min team at Nanjing Normal University, China (https://data.tpdc.ac.cn/zh-hans/data/60dac98d-eec4-41df-9ad5-b1563e5c532c/)	The dataset was primarily creat-ed using deep learning semanti-c segmentation models and mu-lti-source remote sensing image-ry, and it is used for calculati-ng BD
BH data	2020	10m	the Chinese building height estimate at 10 m resolution (CNBH-10 m) dataset (2020)[25], from the GC3S team at Fudan University's School of Life Sciences (https://zenodo.org/records/7827315)	The model was constructed and optimized using a Random F-orest (RF) model, with high si-mulation accuracy (Root Mean Square Error = 6.1 , Mean Err-or = 5.2 , R^2 = 0.77). The roo-ftop data and building height data were used to calculate BHD, and SVF, in order to vis-ually reflect the urban three-di-mensional morphological features

Table 1. Data Sources and Descriptions.

retrieval process was (1) calculate the brightness temperature T_i using the thermal infrared bands of Landsat 8, (2) calculate the surface emissivity ε_i using the red and near-infrared bands of Landsat 8, and (3) retrieve the atmospheric water vapour content using MOD1B bands 2 and 19[26] and calculate the atmospheric transmittance τ_i of MODIS bands 31 and 32 to substitute for the atmospheric transmittance of Landsat 8 bands 10 and 11. The LST was then obtained by calculating those 3 parameters[27,28]. The split-window algorithm is formulated as follows:

$$T_s = A_0 + A_1 T_{10} - A_2 T_{11} \tag{1}$$

where T_s is the surface temperature; T_{10} and T_{11} are the brightness temperatures of Landsat 8 bands 10 and 11; and A_0, A_1, and A_2 are coefficients. The calculation equation for those coefficients is

$$A_0 = \frac{a_{10} D_{11}(1 - C_{10} - D_{10})}{D_{11}C_{10} - D_{10}C_{11}} - \frac{a_{11} D_{10}(1 - C_{11} - D_{11})}{D_{11}C_{10} - D_{10}C_{11}} \tag{2}$$

$$A_1 = 1 + \frac{D_{10}}{D_{11}C_{10} - D_{10}C_{11}} + \frac{b_{10} D_{11}(1 - C_{10} - D_{10})}{D_{11}C_{10} - D_{10}C_{11}} \tag{3}$$

$$A_2 = \frac{D_{10}}{D_{11}C_{10} - D_{10}C_{11}} + \frac{b_{11} D_{10}(1 - C_{11} - D_{11})}{D_{11}C_{10} - D_{10}C_{11}} \tag{4}$$

$$C_i = \varepsilon_i \tau_i \tag{5}$$

$$D_i = (1 - \tau_i)[1 + (1 - \varepsilon_i)\tau_i)] \tag{6}$$

where ε_i is the surface emissivity for band *i*, and τ_i is the atmospheric transmittance for band *i*. Based on the temperature range of the study area, the retrieval regression coefficients for the thermal infrared sensor in various temperature ranges derived by Offer and Qin Zhihao et al.[29] were selected, with $a_{10} = -64.608$, $a_{11} = -69.022$, b_{10}= 0.440, and b_{11}= 0.476.

Driver selection of the land surface temperature

This study aimed to comprehensively explore the formation mechanisms of the UHI effect from both 2D and 3D perspectives. The selected factors covered multiple aspects, including urban spatial morphology, land cover, surface radiation characteristics, and human activities.The meaning of each driving factor is shown in Table 2.

Artificial structures in cities, such as buildings and roads, increase surface roughness, which affects the airflow patterns within the atmospheric boundary layer. In areas with dense buildings, the higher roughness causes heat to accumulate between narrow streets and building clusters, creating an "urban canyon effect" that inhibits the cooling process at night and maintains higher LSTs[30,31]. Additionally, the complex building morphology and terrain structures increase the frequency of solar radiation reflection and refraction, leading to differences in surface radiation energy accumulation across different areas and further exacerbating the spatial heterogeneity of the LST[32]. Based on this, the study selected 4 urban spatial morphology parameters—BH, BD, BHD, and SVF[33]—as 3D driving factors influencing LST.

Land cover characteristics are also key factors influencing the UHI effect. As a major city with over 10 million residents, Tianjin's diverse land surface types have a complex effect on surface temperature. In conjunction with land use data, 4 land surface parameters—the NDVI, the NDWI, the PGS, and the PIS—were selected as driving factors to investigate the relation between land cover characteristics and the UHI effect across various land use types. Albedo, which reflects surface radiation characteristics, was also chosen as a land surface parameter[34,35].

Given that areas with high human activity are often where the UHI effect is most pronounced, the NLI was selected as a socioeconomic parameter to indirectly reflect urban population density, economic activity levels, and energy consumption[36]. Along with the land surface parameters, the NLI was included as one of the 2D driving factors influencing LST.

Extreme gradient boosting

Extreme gradient boosting (XGBoost) is an improved algorithm based on the gradient boosting decision tree algorithm[38]. Integrating multiple classification and regression tree models enhances the model's predictive performance. Relative to other machine learning methods, such as random forests and support vector machines, XGBoost offers significant advantages in terms of training speed and model accuracy[39]. Through iterative learning and regularization control, XGBoost can capture the complex interaction effects between LST driving factors and efficiently handle multi-dimensional, nonlinear, and highly correlated driving factors. Moreover, XGBoost has the capability to automatically handle missing values, simplifying the data preprocessing process. In recent years, some scholars have applied XGBoost to explore the driving factors of LST[40,41].

When analyzing LST, the data often exhibit nonlinear characteristics and significant interaction effects between variables. Additionally, due to the different sources of the driving factors, there may be missing values or scale differences. Therefore, this study selects XGBoost as the baseline model to investigate the driving factors

Type	Driving factor (abbreviation)	Formula	Meaning (30 m × 30 m)
Urban spatial morphology p-arameters (3D perspectiv-es)	Building height (BH)	$\mathrm{BH} = \frac{\sum_{i=1}^{n} h_i}{n}$	Average building heigh-t within the unit
	Building height deviation (BHD)	$\mathrm{BHD} = \sqrt{\frac{\sum_{i=1}^{n} (h_i - \mathrm{BH})^2}{n}}$	Standard deviation of all building heights within the unit
	Building density (BD)	$\mathrm{BD} = \frac{\sum_{i=1}^{n} S_i}{A}$	Proportion of rooftop elements within the unit
	Sky view factor (SVF)	$SVF = 1 - \sum_{i=1}^{n} \sin^2 \beta_i \left(\frac{\alpha_i}{360^\circ}\right)$	Average SVF within the unit
Land surface parameters (2D perspectiv-es)	Normalised difference vegetat-ion index (NDVI)	$\mathrm{NDVI} = \frac{\mathrm{NIR}-\mathrm{RED}}{\mathrm{NIR}+\mathrm{RED}}$	Average NDVI within the unit
	Normalised difference water index (NDWI)	$\mathrm{NDWI} = \frac{\mathrm{GREEN}-\mathrm{NIR}}{\mathrm{GREEN}+\mathrm{NIR}}$	Average NDWI within the unit
	Percentage of impervious surface (PIS)	$\mathrm{PIS} = \frac{A_{IS}}{A}$	Proportion of impervio-us surface within the unit
	Percentage of green space (PGS)	$\mathrm{PGS} = \frac{A_{GS}}{A}$	Whether the unit is gr-een space
	Albedo	$Albedo = 0.356\rho_{TM1} + 0.356\rho_{TM3} + 0.356\rho_{TM4} + 0.356\rho_{TM5} + 0.356\rho_{TM7} - 0.0018$	Average albedo within the unit
Socioeconomic parameters (2D perspectiv-es)	Night light index (NLI)	$\mathrm{NLI} = \frac{\sum_{i=1}^{N} I_i}{N}$	Average NLI within the unit

Table 2. Influencing factors descriptions In this context, n is the number of buildings within the spatial unit; h_i and S_i are the height and base area of building , respectively; A is the area of the spatial unit; NIR, RED, and GREEN represent the near-infrared, red, and green bands of Landsat 8; A_{IS} and A_{GS} are the impervious surface area and green space area within the spatial unit, respectively; ρ_{TM1}, ρ_{TM3}, ρ_{TM4}, ρ_{TM5}, and ρ_{TM7} represent the blue, red, near-infrared, SWIR1, and SWIR2 bands of Landsat 8 TM[37], respectively;I_i is the nighttime light value; N is the number of pixels within the spatial unit.

of LST,ten driving factors are used as input parameters for the XGBoost regression model to predict LST, thereby revealing the factors driving the urban heat island effect.The model architecture is as follows.

The XGBoost model iteratively adds new decision trees to fit the residuals between the predicted and actual values of the previous model while incorporating a regularisation term to reduce overfitting. The loss function is optimised using a second-order Taylor expansion. Assuming the model has *k* decision trees, the model equation is

$$\hat{y}_i = \sum_{i=1}^{k} f_k\left(x_i\right) f_k \in F \tag{7}$$

The objective function is formulated as follows:

$$\mathrm{obj(t)} = \sum\nolimits_{\mathrm{i=1}}^{\mathrm{n}} \mathrm{l(y_i, \widehat{y_i})} + \Omega(\mathrm{f(t)}) \tag{8}$$

$$\Omega\left(f\left(t\right)\right) = \gamma T + 0.5\lambda \sum_{j=1}^{T} \omega_j^2 \tag{9}$$

In Eqs. (7) to (9), $\widehat{y_i}$ is the predicted value for the *i*th pixel, F is the set of all classification and regression trees, x_i is the independent variable for the *i*th pixel in the model, $\mathrm{l(y_i, \widehat{y_i})}$ is the loss function of the target $\mathrm{y_i}$ and its prediction $\widehat{\mathrm{y_i}}$, $\Omega(\mathrm{f(t)})$ is the regularisation term of the objective function, γ is the penalty coefficient controlling the number of leaf nodes, T is the number of leaf nodes, and ω_{j} is the weight of the leaf nodes. The vector norms of T and ω_{j} together determine $\Omega(\mathrm{f(t)})$.

Each research unit in this study used Landsat 8 Operational Land Imager imagery with a spatial resolution of 30 m × 30 m as a regular grid sample. The 2D and 3D parameters of the study units were used as input features for the model. All samples were randomly divided into 80% for the training set and 20% for the test set. Based on fivefold cross-validation, the GridSearchCV function and grid search method were used to fine-tune the regression model's parameters to improve its predictive performance. Referring to previous studies[42], the following parameters were adjusted: The learning rate was set within the range of 0.01, 0.1, and 0.2. A learning rate that was too small might have led to overfitting or local optima, whereas a too-large rate might have caused underfitting. The maximum tree depth (Max_depth) was set to between 6 and 10 with an interval of 2, because a deeper tree was more likely to result in overfitting. The number of trees (n_estimators) was set to between 200 and 400 with an interval of 100, because a larger number would increase the model's complexity and expressive power but also raise the risk of overfitting.

SHapley Additive exPlanations

Machine learning models are often considered "black box" models, because they can provide final results but do not show how input conditions influenced the predicted outcomes[43]. To address this, this study introduced the SHapley Additive exPlanations (SHAP) method to interpret the XGBoost regression model. By calculating Shapley values, SHAP quantifies the contribution of each driving factor to the LST prediction and provides an in-depth analysis of interaction effects, addressing the lack of interpretability in machine learning models[44]. That method, derived from cooperative game theory, was used to quantify the contribution of each driving factor to the dependent variable. The calculation method is

$$\varnothing_i = \sum_{S \subseteq M \setminus \{x_i\}} \frac{|S|!(p-|S|-1)!}{p!} (f_x(S \bigcup \{x_i\}) - f_x(S)) \tag{10}$$

where $\varnothing_i$ is the contribution of variable i, M is the set of all input variables, S is the subset of driving factors used in the model, x is the feature vector of the driving factors for a given instance, p is the number of features of the driving factors, and $f_x(S)$ is the predictive function model for the feature values of the driving factors within the subset S.

Considering the interactions between driving factors, the SHAP interaction values extend the Shapley value to explain the interactive effects between 2 highly correlated driving factors based on the Shapley interaction index from game theory. The calculation method is

$$\varnothing_{i,j} = \sum_{S \subseteq M \setminus \{x_{i,j}\}} \frac{|S|!(p-|S|-2)!}{2(p-1)!} \delta_{i,j}(S) \tag{11}$$

$$\delta_{i,j}(S) = f_x(S \bigcup \{i,j\}) - f_x(S \bigcup \{i\}) - f_x(S \bigcup \{j\}) + f_x(S) \tag{12}$$

where i, j are the feature value vectors of 2 correlated instances of driving factors, and $\delta_{i,j}(S)$ describes the combined effect of the 2 features minus their individual effects in conjunction with the external feature subset S.

The SHAP method provides various data visualisation formats, including driving-factor importance plots, SHAP summary plots, SHAP dependence plots, and SHAP interaction plots. Identifying colours in the plots makes it possible to visually distinguish the strength and direction of each driving factor's effect on LST and capture interactions between factors. In this study, SHAP summary and dependence plots of interaction values were selected to analyse the relation between driving factors and LST.The combination of the XGBoost model and SHAP method not only enhances the reliability and interpretability of the model results through detailed feature importance analysis but also comprehensively and accurately reveals the driving mechanisms of LST, providing scientific support for mitigating the urban heat island effect.

Results

Spatial distribution characteristics of driving factors

Figure 3a shows the spatial distribution of LSTs retrieved using the split-window algorithm at 10:47 AM local time on August 28, 2020. The highest temperature was 44.3 °C, and the lowest was 21.9 °C. To validate the accuracy of the retrieved data, the MODIS LST (scene ID:MYD11A2.A2020241.h27v05.061.2021014172341) product around 10:30 AM (local time) on the same day was selected. Pearson correlation analysis yielded $r = 0.96$, validating the accuracy of the retrieved data.It can be seen that the LST gradually decreased as the distance from the central districts increased. Areas in the central districts, where buildings and population were densely concentrated, had higher and more uniform temperatures, whereas the 4 peri-urban districts had generally lower temperatures. However, several regions had extremely high surface temperatures in the Beichen District and the southeastern part of the study area. Areas with greater green space and water body coverage tended to have lower temperatures overall.

Figures 3b–k show the spatial distribution of the various driving factors. It can be seen that vegetation coverage had an overall distribution pattern of "lower in the central area and higher in the surrounding areas". Nighttime light brightness was most concentrated and brightest in the central districts, whereas in the 4 peri-urban districts, it was concentrated in only the central parts of each district, with brightness decreasing as the distances from the centres increased. The central districts had a higher population and BD and a greater proportion of impervious surfaces than the surrounding areas. However, because surface materials in the central districts were primarily dark-coloured and low-reflectivity materials such as concrete and asphalt, the surface albedo was lower than that of the 4 peri-urban districts.

Extreme gradient boosting model representation

Using XGBoost regression, models were established to analyse the relation between driving factors and LST in both the central districts and the 4 peri-urban districts. The model accuracy was evaluated to verify the effectiveness of the XGBoost model. The results are shown in Table 3. The low MAE values and high R^2 values indicated that the model fit the dataset well and provided accurate predictions.

From the model evaluation results, different fitting performances could be observed between the central and 4 peri-urban districts (see Fig. 4). In the central districts, the training set R^2 was as high as 0.95, indicating that the model could explain most of the variation in LSTs within that area. However, the R^2 for the test set dropped to 0.80, suggesting that the model's predictive capability for new data decreased. In contrast, the training and test sets' R^2 for the 4 peri-urban districts were 0.94 and 0.86 respectively, demonstrating better generalisation ability. One reason for this is that, compared to the central districts, the 4 peri-urban districts had a less concentrated distribution of individual factors, allowing the model to better adapt to varying environmental conditions in that region.

The comparison revealed differences in LST behaviour between the central districts and the 4 peri-urban districts, implying that variations in urban function, geographic environment, and economic activity across regions had different levels of effect on the LST. Considering those factors, this study further used the SHAP method to conduct spatial contribution analyses of the driving factors and LST relationship models for both the central districts and the 4 peri-urban districts.

Contribution of driving factors to the land surface temperature

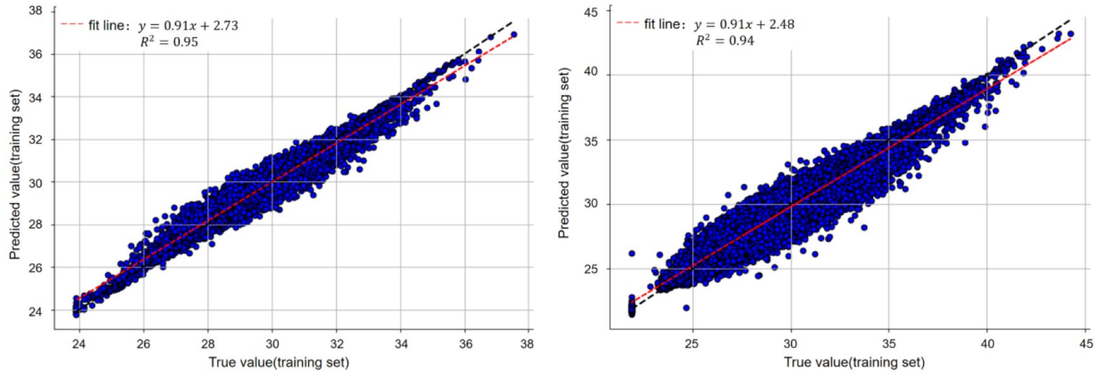
The contribution of driving factors is an indicator that measures how they influence LSTs under specific conditions. In this study, we applied the results of the XGBoost model, optimised through parameter tuning, to the SHAP model to analyse the specific effect of each driving factor on LST.

Figure 5 shows the SHAP summary plots for both the central districts and the 4 peri-urban districts. Panels (**a**) and (**c**) illustrate the contribution of each driving factor to LST, arranged in descending order of importance. In both the central and 4 peri-urban districts, NDVI is the driving factor with the highest explanatory power for LST, with contributions as high as 0.89 and 0.90 respectively; PGS has the lowest contribution, at 0 and 0.02 respectively. That might be because PGS, as a quantification of green space area, does not adequately capture vegetation quality or cooling effects[45].

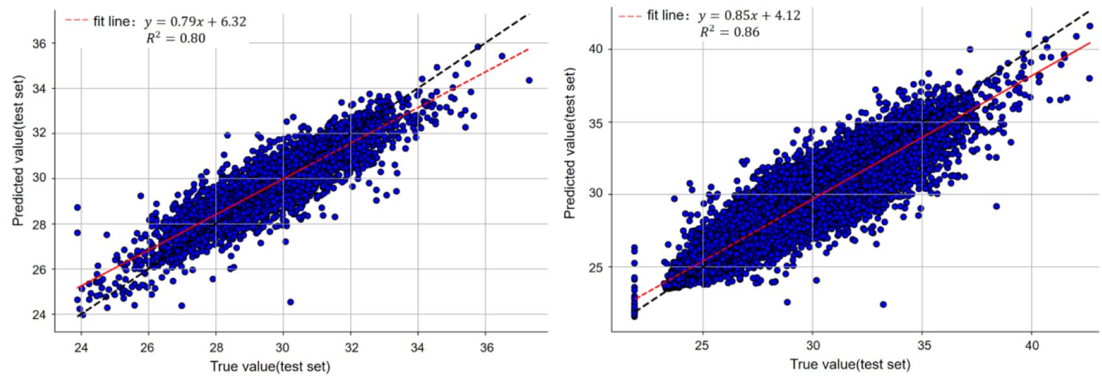
In the central districts, the driving factors ranked second to fifth in terms of contribution were BH (0.30), albedo (0.25), BD (0.22), and the NLI (0.18). The total contributions of land surface parameters and urban spatial morphology parameters were 56.6% and 33.4% respectively, while socioeconomic parameters contributed 8%.

Evaluation metric	Central districts		Four peri-urban districts	
	Training set	Test set	Training set	Test set
R^2	0.95	0.80	0.94	0.86
Mean absolute error	0.28	0.60	0.44	0.67

Table 3. XGBoost model evaluation results.



(**a**) True vs predicted values of the training set



(**b**) True vs predicted values of the test set

Fig. 4. Scatterplots of XGBoost model (left: central districts; right: 4 peri-urban districts).

For the 4 peri-urban districts, the driving factors ranked second to fifth were albedo (0.60), the NLI (0.42), PIS (0.30), and BH (0.24). The total contributions of land surface parameters and urban spatial morphology parameters were 65.7% and 20.3% respectively, while socioeconomic parameters contributed 13.9%.

A comparative analysis found that 2D parameters had a more significant influence on LST than 3D parameters. In the 4 peri-urban districts, the effect of urban spatial morphology parameters on LST was less than in the central districts, while the influence of socioeconomic parameters was greater.

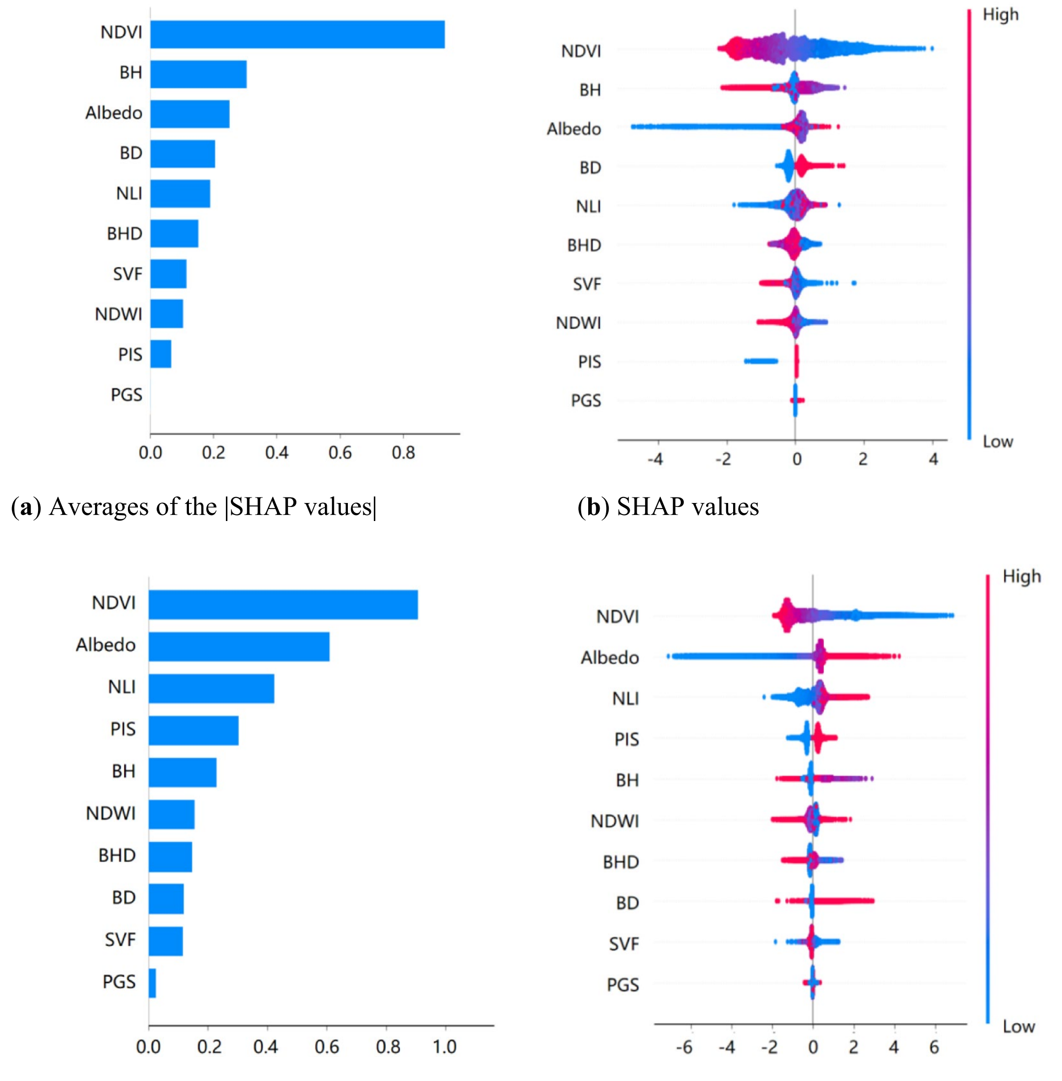
Panels (**b**) and (**d**) in Fig. 5 are summary plots of the SHAP values for each sample, along with the distribution of their feature values. The colours indicate the relation between driving factors and LST, where blue points represent samples with lower feature values, and red points represent samples with higher feature values. From the plots, it can be observed that NDVI is blue in the region where SHAP values exceeded 0, indicating a negative correlation between the NDVI and LST. Albedo also significantly affected LST, with blue points appearing in the region where SHAP values were less than 0, suggesting that lower albedo values suppressed LST. The NLI and PIS were both positively correlated with LST. The PGS sample points are clustered around a SHAP value of 0, further indicating that the PGS had little effect on LST.

Regarding the 4 urban spatial morphology parameters, in the central districts, LST tended to increase as BH, BHD, and the SVF decreased and BD increased. A lower SVF indicates a smaller field of view, implying higher building height and density, reduced air circulation within the city, and slower heat dissipation[46], which led to an increase in LST. In the 4 peri-urban districts, the influences of BH, BHD, and the SVF on LST were consistent with those observed in the central districts, whereas the effects of BD and the NDWI on LST remained unclear.

Interaction between driving factors

To further clarify the mechanisms by which driving factors influenced LST, the 5 driving factors with the highest contributions were selected using the XGBoost model for both the central and 4 peri-urban districts. Those factors were the NDVI, BH, albedo, BD, and the NLI for the central districts, and the NDVI, albedo, the NLI, PIS, and BH for the 4 peri-urban districts. Subsequently, SHAP dependence plots were used to analyse the interaction effects among those driving factors. In the SHAP dependence plots, the x axis represents the value of the driving factors, and the y axis represents the SHAP value of that factor (indicating the extent of its effect on LST). The results are shown in Figs. 6 and 7.

It can be observed that the spatial effect trends of driving factors on LST had both similarities and differences. For the top 5 driving factors, the overall response of LST to the same factor showed spatial consistency (see Fig. 6).That consistency provided a theoretical basis for subsequent regional analyses. In contrast, the interactions between BHD and the NDVI, and between the NDWI and BH, had different effect patterns in the central districts



(**a**) Averages of the |SHAP values| (**b**) SHAP values

(**c**) Averages of the |SHAP values| (**d**) SHAP values

Fig. 5. SHAP summary plots of the study areas (top: central districts; bottom: 4 peri-urban districts).

and the 4 peri-urban districts, as shown in Fig. 7. That reveals that interaction effects had specific patterns in different regions.

Spatial regularity of driving factors affecting land surface temperatures

As shown in Figs. 6a1 to a10 and b1 to b10, the overall spatial response trends of LST to the same driving factors in the central districts and the 4 peri-urban districts were generally similar. For example, when albedo was less than 0.08, both regions showed a significant cooling effect. That phenomenon contrasted with the findings of Liu et al.[22], which might be attributed to a high heat capacity and high reflectivity of buildings in the central districts that impeded heat dissipation. In contrast, in the 4 peri-urban districts, the effects of vegetation evapotranspiration effectively reduced the LST.

As the NDVI increased, LSTs in both the central and 4 peri-urban districts showed a concave downward trend. When the NDVI reached a certain value, LST significantly decreased. As shown in Figs. 6 a6 and 6 b6, those 2 driving factors correlated notably. Part of the solar radiation absorbed by vegetation was converted into chemical energy through photosynthesis, while another part was absorbed from the environment through transpiration[47]. Meanwhile, an increase in albedo led to a higher reflection of solar radiation from the surface, reducing the solar radiation absorbed by the underlying surface[48]. The interaction of those 2 factors can effectively reduce LST.

Spatial variability of driving factors affecting land surface temperatures

Although the overall response trends of LSTs to driving factors in the central and 4 peri-urban districts were similar, there were still significant spatial differences between the 2 regions. That was particularly evident in the

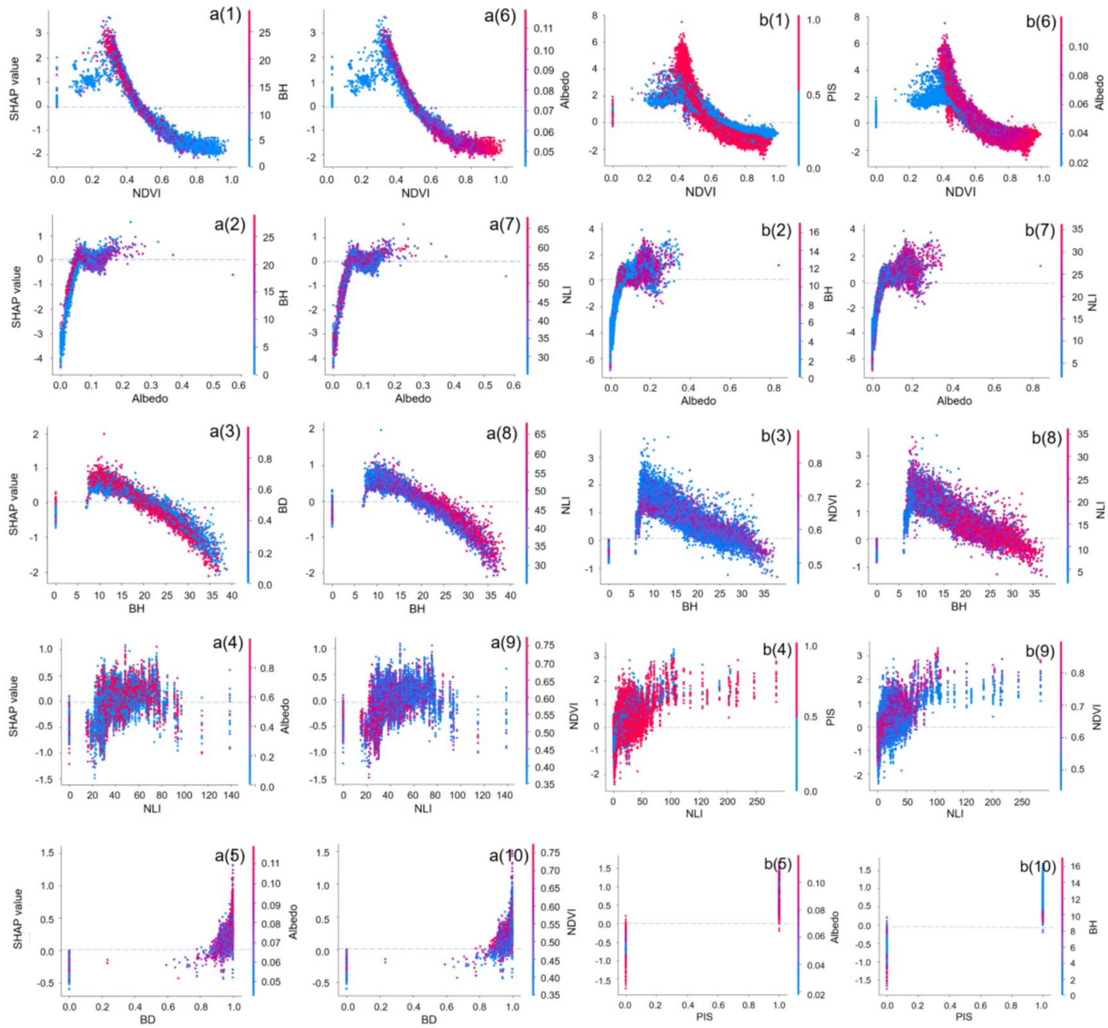


Fig. 6. Partial dependence plots of the first 5 characteristics affecting LST. The *x* axes represent the value of the driving factors, and the *y* axes represent the SHAP value of that factor. (**a**) Central districts. (**b**) Four peri-urban districts.

interaction effects of the BHD_NDVI and NDWI_BH indicators, where the influence patterns differed markedly between the 2 regions (see Fig. 7).

For example, the interaction of BHD_NDVI in the central districts was more complex. When the BHD was in the range of 7.5 to 11.5 m, an increase in the NDVI significantly reduced LST, but outside that range, whether the BHD increased or decreased, LST rose. In contrast, in the 4 peri-urban districts, the interaction between the BHD and the NDVI was weaker, with only a low NDVI significantly raising LST when the BHD was at 4 m. Additionally, the interaction effect of NDWI_BH in the central districts showed that a decrease in BH enhanced the main effect of the NDWI, leading to an increase in LST. However, in the 4 peri-urban districts, when the NDWI reached 0.5, a decrease in BH helped lower LST. That indicated that when the water body index reached a certain threshold, changes in BH could significantly influence the temperature in the 4 peri-urban districts. In contrast, when the NDWI was not equal to 0.5, any changes in BH increased LST. That further demonstrated that the mechanisms by which driving factors affected LST differed across regions.

Land surface temperature response to factor interactions

Specifically, the response of LST to driving factors in the central districts was closely related to urban structure and land use. For example, when albedo was less than 0.08 or BH was greater than 20 m, a reduction in BH and BD resulted in a cooling effect (Figs. 6a2 and 6a3). However, when the NDVI was less than 0.45, a decrease in albedo (below 0.07) led to an increase in LST (Fig. 6a6). Figure 6a3 shows that the negative correlation between BH and BD (high BH and low BD) brought about a significant cooling effect, and the interaction between BH and BD had a stronger influence on LST than the interaction between the NLI and the NDVI (Fig. 6a9). That was mainly because the central districts were densely built with widespread impervious surfaces, where urban morphology factors like BD and BH significantly affected the LST[49]. It can be concluded that the urban morphology and surface characteristics of the central districts played a crucial role in LST variation, especially

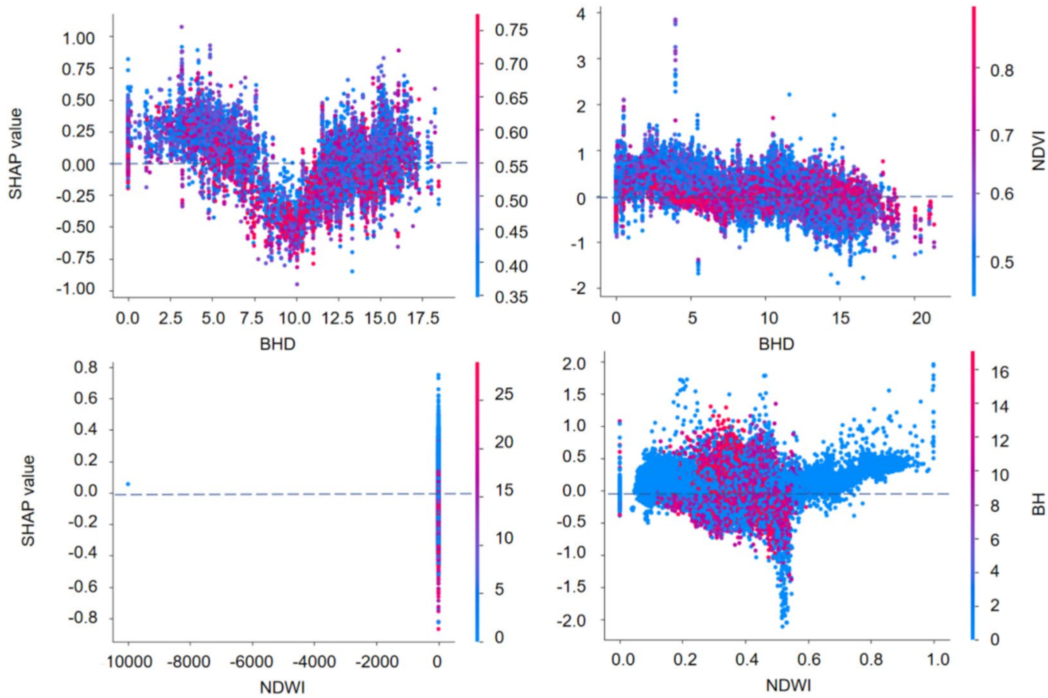


Fig. 7. Partial dependence plots of BHD_NDVI and NDWI_BH. The *x* axes represent the value of the driving factors, and the *y* axes represent the SHAP value of that factor. (Left: central districts; right: 4 peri-urban districts.)

in highly dense urban environments, where the cooling effect depended on the synergistic interaction between buildings and vegetation.

In contrast, the response of LST to driving factors in the 4 peri-urban districts was more closely related to natural environmental and land use characteristics. For example, when the NDVI was less than 0.5, an increase in the PIS led to a rise in LST (Fig. 6b1). Before BH exceeded 25 m, low BH and low NLI values contributed significantly to an increase in LST, but once BH exceeded 25 m, the interaction between BH and NLI started to suppress the LST (Fig. 6b8). Figures 6a9 and 6b9 show that the interaction values of the NLI increased significantly in the 4 peri-urban districts, indicating a stronger response to LST. That was because, although the 4 peri-urban districts had ample green space, urban expansion and the increase in impervious surfaces, coupled with intensified land use diversity and economic activities, had altered the heat absorption and release patterns, making LST more sensitive to the NLI[50]. Figure 6b1 shows that the interaction between 2D parameters was most pronounced in areas with dense natural surfaces and low human activity and BDs. As the PIS increased, a continuous rise in the NDVI significantly reduced LSTs. That might have been due to industrial parks and infrastructure materials (e.g., asphalt, concrete) having high heat capacities and thermal conductivity, which enabled rapid absorption and storage of solar radiation, while the additional heat generated by industrial activities further raised the LST[51,52]. However, vegetation's transpiration and shading effects effectively mitigated that warming effect.

In summary, the differing levels of development led to distinct mechanisms by which driving factors influenced LST variations in the central and 4 peri-urban districts. The LST changes in the central districts were significantly influenced by the synergistic interaction between buildings and vegetation, while in the 4 peri-urban districts, the LST was more affected by the combined effects of the natural environment and urban expansion. Rational planning of green spaces and infrastructure is essential for temperature regulation in each region.

Discussion

The study found that the spatial distribution of LSTs was closely related to urban morphology and development planning. As shown in Fig. 3a, local high-temperature areas were distributed mainly in the industrial parks, technology parks, and industrial zones located in the Beichen, Donglit, and Jinnan districts, including the Dongli aviation, Binhai heavy machinery, and Tianjin Shuanggang industrial zones. Those areas contain high-density factories and warehouses constructed with materials of high heat capacity and thermal conductivity. Additionally, industrial production, especially in heavy industries and chemical and pharmaceutical industries, generates extra heat[53], contributing to the rise in LST in those regions. In contrast, low-temperature areas are concentrated in regions with high vegetation coverage, such as reservoirs, rivers, wetland parks, and farmland, as well as in the northern parts of the Xiqing and Beichen districts. As a demonstration area for ecological civilisation, Xiqing District benefits from green spaces and open areas that effectively regulate LSTs, while Beichen District, an eco-liveable city, also mitigates heat through its substantial green spaces and water bodies[54].

Thus, it can be seen that high-temperature areas are closely related to high BD, population density, and industrial zones, while low-temperature areas are associated with vegetation coverage and water bodies. That indicates that LSTs exhibit significant spatial heterogeneity, depending on the level of urban development and functional zoning.

The SHAP explanation of the XGBoost model revealed that, in the central districts, land surface parameters were the most influential, with NDVI having the highest contribution and a negative correlation with LST, followed by albedo, which was positively correlated. In the peri-urban districts, socioeconomic parameters had a greater contribution than in the central districts, while the impact of urban morphology parameters decreased significantly. In both regions, the influence of 2D parameters on LST was much greater than that of 3D parameters (Fig. 6). This suggests that, in regulating the urban thermal environment, 2D land cover and socioeconomic factors are more influential than 3D urban morphology parameters. This finding aligns with Chen et al.[18], who found that the percentage of building coverage at the 2D level had the greatest impact on surface temperature in spring and summer. However, Yuan et al.[55] concluded that 3D building indicators had a stronger effect on LST changes. This discrepancy may result from differences in urban morphology and development patterns, as well as the spatial distribution of input data affecting model outcomes. Additionally, the 3D urban morphology parameters might have been constrained by factors such as scanning angles and geographical location[56].

Liu et al.[57] pointed out that the relationship between urban morphology indicators (such as Building Coverage Ratio, BCR) and LST exhibits a complex nonlinear pattern, especially at certain threshold points, where the influence on LST reverses when these indicators exceed a certain limit. Our findings are consistent with this. In terms of the impact of pairwise factor interactions on LST, the NDVI_PIS combination in the peri-urban districts has the highest response to LST (Fig. 6b1), where an increase in PIS leads to a rise in LST when NDVI < 0.5. In contrast, in the central districts, the BH_BD combination has a stronger impact (Fig. 6a3), where a decrease in BD values leads to a significant cooling effect when BH > 20 m, and the interaction between BH and BD has a stronger influence on LST than the NLI_NDVI combination (Fig. 6a9). This suggests that in high-density building areas, the synergistic effect of 3D urban morphology parameters plays an important role in regulating the thermal environment. This study not only continues the research of Liu et al.[57] on the nonlinear relationship between influencing factors and LST, but also incorporates socioeconomic factors to further clarify the different mechanisms of 2D and 3D urban morphology parameters. By quantifying the interactions between different factors, this study not only confirms the significant impact of 2D parameters on LST but also provides a multi-factor-based thermal environment regulation strategy for future urban planning.

Based on the above discussion, the interaction between driving factors significantly affects LST and its spatial effects. Therefore, urban planning and land use should focus on the coordinated adjustment of multiple factors, and local conditions should be considered to address the spatial differences in LSTs within a city. In densely built and developed central districts, vertical greening can be used to enhance building shading effects. In the 4 peri-urban districts, where natural surface coverage is higher, industrial land expansion in the northwest should be limited, maintaining high water and vegetation coverage, while in the southeast, BHs should be controlled and building layouts optimised. For industrial parks, increasing green spaces and vegetation, promoting green building designs, using high-reflectivity materials, or implementing roof and wall greening to enhance evapotranspiration could help reduce heat absorption. Additionally, optimising industrial production processes, using energy-efficient equipment, and reducing energy consumption and industrial heat emissions can further alleviate urban heat problems[54]. These measures can effectively regulate the urban thermal environment, enhance the city's climate resilience, and promote the sustainable development of Tianjin, and the proposed regional heat environment regulation strategies are also of great reference value for rapidly developing cities.

Additionally, this study had some limitations: First, it did not analyse the effect mechanisms of LSTs across temporal dimensions, such as diurnal and seasonal variations. Second, the study was conducted at a 30 × 30 m scale, and previous research has shown that the correlation between LST and driving factors might vary across different scales. Further research is required to conduct a more detailed comparative analysis at the urban block scale, focusing on the effect of urban morphology on LST and its temporal dynamics.

Conclusions

This study reveals significant differences in the UHI mechanism between the highly urbanized central districts and the rapidly developing peri-urban districts. By combining XGBoost and SHAP methods, the contributions of various driving factors to LST were analyzed, and the similarities and differences in UHI mechanisms between these two areas were compared. The main findings can be summarized as follows.

The regulatory effect of 2D parameters on LST is greater than that of 3D parameters. In both the central districts and the peri-urban districts, the driving factor with the highest contribution to LST was NDVI, BH in the central districts and Albedo in the peri-urban areas ranked second in contribution. Compared to the central districts, the impact of urban spatial morphology parameters on LST in the peri-urban districts decreased, while the influence of socioeconomic parameters increased. This indicates that land surface parameters and socioeconomic factors at the 2D scale play a more critical role in regulating the UHI effect.

Regarding the interaction of pairwise factors on LST, the effects in the central districts and the peri-urban districts differ. In the central districts, the interaction between BH and BD had the greatest impact on LST. When BH exceeds 20 m, a reduction in BD significantly cools the temperature, indicating that in high-density built environments, the synergistic effect of urban morphology is crucial for regulating the UHI effect. In contrast, in the peri-urban districts, the interaction between NDVI and PIS had the greatest impact on LST, particularly when NDVI was below 0.5, where an increase in PIS led to an increase in LST. This suggests that the regulatory mechanisms of the UHI effect differ significantly at different stages of urban development, with differences in urban morphology and land use patterns leading to varying UHI outcomes.

When formulating heat island mitigation strategies for different regions, regionalized thermal environment control measures should be developed based on the specific development stage and spatial characteristics of the city.For densely built, highly developed urban areas, the focus should be on optimising 2D parameters to reduce LSTs. In rapidly developing areas with dense natural surfaces and low human activity and BD, socioeconomic factors must be comprehensively considered to effectively control LSTs.Future research could explore the regulatory mechanisms of the urban heat island effect at different spatial scales and time dimensions, such as diurnal and seasonal variations.

Data availability
All data used during this study are included in this article.

Received: 23 October 2024; Accepted: 23 May 2025
Published online: 28 May 2025

References
1. Oke, T. R..The energetic basis of the urban heat island.*Quarterly journal of the royal meteorological society*,**108(455)**,1–24(1982).
2. Bokaie, M.,Zarkesh, M. K.,Arasteh, P. D.,&Hosseini, A.Assessment of urban heat island based on the relationship between land surface temperature and land use/land cover in Tehran.*Sustainable Cities and Society*,**23**,94–104.: https://doi.org/10.1016/j.scs.2016.03.009 (2016).
3. Ding, C. Building height restrictions,land development and economic costs.*Land use policy*,**30(1)**,485–495. https://doi.org/10.1016/j.landusepol.2012.04.016 (2013).
4. Talukdar, S., Singha, P., Mahato, S., Praveen, B. & Rahman, A. Dynamics of ecosystem services (ESs) in response to land use land cover (LU/LC) changes in the lower Gangetic plain of India. *Ecol. Ind.* **112**, 106121. https://doi.org/10.1016/j.ecolind.2020.106121 (2020).
5. Deilami, K.,Kamruzzaman, M.,&Liu, Y. Urban heat island effect: A systematic review of spatio-temporal factors, data, methods, and mitigation measures. *International journal of applied earth observation and geoinformation*,**67**,30–42. https://doi.org/10.1016/j.jag.2017.12.009 (2018).
6. Hulley, G. C.,Ghent, D.,Göttsche, F. M.,Guillevic, P. C.,Mildrexler, D. J.,&Coll, C. Land surface temperature. *In Taking the Temperature of the Earth*,pp,57–127. Elsevier. https://doi.org/10.1016/B978-0-12-814458-9.00003-4 (2019).
7. Fitria, R.,Kim, D.,Baik, J., Choi, M. Impact of biophysical mechanisms on urban heat island associated with climate variation and urban morphology. *Scientific Reports*,**9(1)**,19503. https://doi.org/10.1038/s41598-019-55847-8 (2019).
8. Petralli, M.,Massetti, L.,Brandani, G.,&Orlandini, S. Urban planning indicators: useful tools to measure the effect of urbanization and vegetation on summer air temperatures.*International Journal of Climatology*,**34(4)**. https://doi.org/10.1002/joc.3760 (2014).
9. Mabon, L.,&Shih, W. Y. Mapping the socio-political landscape of heat mitigation through urban greenspaces: the case of Taipei Metropolis.*Environment and urbanization*,**31(2)**,552–574. https://doi.org/10.1177/0956247818767318 (2019).
10. Koc, M.,&Acar, A. Investigation of urban climates and built environment relations by using machine learning.*Urban Climate*,**37**,100820. : https://doi.org/10.1016/j.uclim.2021.100820 (2021).
11. Guo, G.,Wu, Z.,&Chen, Y. Complex mechanisms linking land surface temperature to greenspace spatial patterns: Evidence from four southeastern Chinese cities.*Science of the total Environment*,**674**,77–87. https://doi.org/10.1016/j.scitotenv.2019.03.402 (2019).
12. Effati, F., Karimi, H. & Yavari, A. Investigating effects of land use and land cover patterns on land surface temperature using landscape metrics in the city of Tehran. *Iran. Arabian Journal of Geosciences* **14**(13), 1240. https://doi.org/10.1007/s12517-021-07433-4 (2021).
13. Yan L, Jia W, Zhao S. The cooling effect of urban green spaces in metacities: A case study of Beijing, China's capital. *Remote Sensing*,**13(22)**,4601.: https://doi.org/10.3390/rs13224601 (2021).
14. Sahak, A. S., Karsli, F., Gormus, E. T. & Ahmadi, K. Seasonal monitoring of urban heat island based on the relationship between land surface temperature and land use/cover: a case study of Kabul City. *Afghanistan. Earth Science Informatics* **16**(1), 845–861. https://doi.org/10.1007/s12145-022-00918-0 (2023).
15. Zhou, D.,Zhao, S.,Zhang, L.,&Liu, S. Remotely sensed assessment of urbanization effects on vegetation phenology in China's 32 major cities.*Remote Sensing of Environment*,**176**,272–281.: https://doi.org/10.1016/j.rse.2016.02.010 (2016).
16. Scarano, M.,&Mancini, F. Assessing the relationship between sky view factor and land surface temperature to the spatial resolution.*International Journal of Remote Sensing*,**38(23)**,6910–6929.: https://doi.org/10.1080/01431161.2017.1368099 (2017).
17. Berger, C.,Rosentreter, J.,Voltersen, M.,Baumgart, C.,Schmullius, C.,&Hese, S. Spatio-temporal analysis of the relationship between 2D/3D urban site characteristics and land surface temperature.*Remote sensing of environment*,**193**,225–243.: https://doi.org/10.1016/j.rse.2017.02.020 (2017).
18. Chen, J.,Zhan, W.,Du, P.,Li, L.,Li, J.,Liu, Z., &Xia, J. Seasonally disparate responses of surface thermal environment to 2D/3D urban morphology.*Building and Environment*,**214**,108928. : https://doi.org/10.1016/j.buildenv.2022.108928 (2022).
19. Gupta K, Garg P, Roy A. Assessment of spatio-temporal and diurnal urban heat island intensities in Delhi urban agglomeration using a high resolution weather research and forecasting model[C]//AGU Fall Meeting Abstracts. 2021, 2021: GH13A-02.
20. Chen, Y.,Xie, M.,Chen, B.,Wang, H.,&Teng, Y.,2023.Surface regional heat (cool) island effect and its diurnal differences in arid and semiarid resource-based urban agglomerations. *Chinese Geographical Science*,**33(1)**,131–143. https://doi.org/10.1007/s11769-022-1324-y (2023).
21. Xiong Y, Zhang F. Effect of human settlements on urban thermal environment and factor analysis based on multi-source data: A case study of Changsha city. *Journal of Geographical Sciences*, **31(6),**819–838. https://doi.org/10.1007/s11442-021-1873-5(2021).
22. Liu, Y.,Xu, Y.,Zhang, Y.,Han, X.,Weng, F.,Xuan, C.,&Shu, W. Impacts of the urban spatial landscape in Beijing on surface and canopy urban heat islands.*Journal of Meteorological Research*,**36(6)**,882–899. https://doi.org/10.1007/s13351-022-2045-y (2022).
23. Yang, J. & Huang, X. 30 m annual land cover and its dynamics in China from 1990 to 2019. *Earth System Science Data Discussions* **2021**, 1–29. https://doi.org/10.5194/essd-13-3907-2021 (2021).
24. Zhang, Z.,Qian, Z.,Zhong, T.,Chen, M.,Zhang, K.,Yang, Y., ... &Yan, J. Vectorized rooftop area data for 90 cities in China.*Scientific Data*,**9(1)**, 66. https://doi.org/10.1038/s41597-022-01168-x (2022).
25. Wu, W. B.,Ma, J.,Banzhaf, E.,Meadows, M. E.,Yu, Z. W.,Guo, F. X., ...&Zhao, B. A first Chinese building height estimate at 10 m resolution (CNBH-10 m) using multi-source earth observations and machine learning.*Remote Sensing of Environment*,**291**,113578.: https://doi.org/10.1016/j.rse.2023.113578 (2023).
26. Qin, Z. H.,Li, W. J.,Xu, B.,Chen, Z. X.,&Liu, J. The estimation of land surface emissivity for Landsat TM6.*Remote Sens. Land Resour*,**3**,28–32 (2004).
27. Dwivedi, A.,&Khire, M. V. Application of split-window algorithm to study Urban Heat Island effect in Mumbai through land surface temperature approach.*Sustainable cities and Society*,**41**,865–877. : https://doi.org/10.1016/j.scs.2018.02.030 (2018).

28. Wu, L.,&Yao, K. Land surface temperature retrieval of Landsat 8 images based on split window algorithm.*Power Systems And Big Data*,**21(4)**,18–25 (2018).
29. Rozenstein, O.,Qin, Z.,Derimian, Y.,&Karnieli, A. Derivation of land surface temperature for Landsat-8 TIRS using a split window algorithm.*Sensors*,**14(4)**,5768–5780. : https://doi.org/10.3390/s140405768 (2014).
30. Miao C, Yu S, Hu Y, et al. Review of methods used to estimate the sky view factor in urban street canyons. *Building and environment*,**168**,106497.: https://doi.org/10.1016/j.buildenv.2019.106497 (2020).
31. Zhou R, Xu H, Zhang H, et al. Quantifying the relationship between 2D/3D building patterns and land surface temperature: Study on the metropolitan Shanghai. *Remote Sensing*,**14(16)**,4098.: https://doi.org/10.3390/rs14164098 (2022).
32. Yang J, Yang Y, Sun D, et al. Influence of urban morphological characteristics on thermal environment. *Sustainable Cities and Society*,**72**,103045.: https://doi.org/10.1016/j.scs.2021.103045 (2021).
33. Guo G, Zhou X, Wu Z, et al. Characterizing the impact of urban morphology heterogeneity on land surface temperature in Guangzhou, China. *Environmental Modelling & Software*,**84**,427–439. : https://doi.org/10.1016/j.envsoft.2016.06.021 (2016).
34. Zhang X, Jiao Z, Zhao C, et al. Review of land surface albedo: variance characteristics, climate effect and management strategy. *Remote Sensing*, **14(6)**,1382.: https://doi.org/10.3390/rs14061382 (2022).
35. Liang, S.,Shuey, C. J.,Russ, A. L.,Fang, H.,Chen, M.,Walthall, C. L., ...&Hunt Jr, R. Narrowband to broadband conversions of land surface albedo: II. Validation.*Remote Sensing of Environment*,**84(1)**,25–41. https://doi.org/10.1016/S0034-4257(02)00068-8 (2003).
36. Chen, S.,Yu, Z.,Liu, M.,Da, L.,&Faiz ul Hassan, M. Trends of the contributions of biophysical (climate) and socioeconomic elements to regional heat islands.*Scientific reports*,**11(1)**,12696. https://doi.org/10.1038/s41598-021-92271-3 (2021).
37. Liang,S.Narrowband to broadband conversions of land surface albedo I:Algorithms[J].*Remote sensing of environment*,**76(2)**,213–238.: https://doi.org/10.1016/S0034-4257(00)00205-4(2001).
38. Chen, T.,&Guestrin, C. August.Xgboost: A scalable tree boosting system. *In Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*,pp.785–794 (2016).
39. Fan, J.,Wang, X.,Wu, L.,Zhou, H.,Zhang, F.,Yu, X., ... &Xiang, Y. Comparison of Support Vector Machine and Extreme Gradient Boosting for predicting daily global solar radiation using temperature and precipitation in humid subtropical climates: A case study in China.*Energy conversion and management*,**164**,102–111. https://doi.org/10.1016/j.enconman.2018.02.087 (2018).
40. Sun, Y., Gao, C., Li, J., Wang, R. & Liu, J. Evaluating urban heat island intensity and its associated determinants of towns and cities continuum in the Yangtze River Delta Urban Agglomerations. *Sustain. Cities Soc.* **50**, 101659. https://doi.org/10.1016/j.scs.2019.101659 (2019).
41. Zhou, L.,Hu, F.,Wang, B.,Wei, C.,Sun, D.,&Wang, S. Relationship between urban landscape structure and land surface temperature: Spatial hierarchy and interaction effects.*Sustainable Cities and Society*,**80**,103795. https://doi.org/10.1016/j.scs.2022.103795 (2022).
42. Yu, S.,Chen, Z.,Yu, B.,Wang, L.,Wu, B.,Wu, J.,&Zhao, F. Exploring the relationship between 2D/3D landscape pattern and land surface temperature based on explainable eXtreme Gradient Boosting tree: A case study of Shanghai, China.*Science of the Total Environment*,**725**,138229.: https://doi.org/10.1016/j.scitotenv.2020.138229 (2020).
43. Lundberg, S. A unified approach to interpreting model predictions. (2017).
44. Štrumbelj, E.,&Kononenko, I. Explaining prediction models and individual predictions with feature contributions. *Knowledge and information systems*,**41**,647–665. https://doi.org/10.1007/s10115-013-0679-x (2014).
45. Zhou, W.,Yu, W.,Qian, Y.,Han, L.,Pickett, S. T.,Wang, J., ...&Ouyang, Z. Beyond city expansion: multi-scale environmental impacts of urban megaregion formation in China.*National Science Review*,**9(1)**,nwab107. https://doi.org/10.1093/nsr/nwab107 (2022).
46. Yang J, Yang Y, Sun D, et al. Influence of urban morphological characteristics on thermal environment. *Sustainable Cities and Society*, **72**,103045.: https://doi.org/10.1016/j.scs.2021.103045 (2021).
47. Wu, X.,Li, B.,Li, M.,Guo, M.,Zang, S.,&Zhang, S. Examining the relationship between spatial configurations of urban impervious surfaces and land surface temperature.*Chinese Geographical Science*,**29**,568–578. https://doi.org/10.1007/s11769-019-1055-x (2019).
48. Zhao, C.,Fu, G.,Liu, X.,&Fu, F. Urban planning indicators, morphology and climate indicators: A case study for a north-south transect of Beijing, China.*Building and Environment*,**46(5)**,1174–1183. https://doi.org/10.1016/j.buildenv.2010.12.009 (2011).
49. Sun, W.,&Li, T. Building height trends and their influencing factors under China's rapid urbanization: A case study of Guangzhou, 1960–2017.*Chinese Geographical Science*,**30**,993–1004. https://doi.org/10.1007/s11769-020-1162-8 (2020).
50. Weng, Q. Thermal infrared remote sensing for urban climate and environmental studies: Methods, applications, and trends.*ISPRS Journal of photogrammetry and remote sensing*,**64(4)**,335–344.: https://doi.org/10.1016/j.isprsjprs.2009.03.007 (2009).
51. Klok, L.,Zwart, S.,Verhagen, H.,&Mauri, E. The surface heat island of Rotterdam and its relationship with urban surface characteristics.*Resources, conservation and recycling*,**64**,23–29. https://doi.org/10.1016/j.resconrec.2012.01.009 (2012).
52. Zhou, B.,Rybski, D.,&Kropp, J. P. The role of city size and urban form in the surface urban heat island.*Scientific reports*,**7(1)**,4791. https://doi.org/10.1038/s41598-017-04242-2 (2017).
53. Cong, Z.,Shen, Q.,Zhou, L.,Sun, T., &Liu, J. Evapotranspiration estimation considering anthropogenic heat based on remote sensing in urban area.*Science China Earth Sciences*,**60**,659–671. https://doi.org/10.1007/s11430-016-0216-3 (2017).
54. Degefu, M. A., Argaw, M., Feyisa, G. L. & Degefa, S. Effects of urbanization on the relationship between greenspace patterns and evolution of regional heat island in cities of Ethiopia. *Chinese Journal of Population, Resources and Environment* **19**(4), 330–343. https://doi.org/10.1016/j.cjpre.2022.01.006 (2021).
55. Yuan, B.,Zhou, L.,Hu, F.,&Wei, C.,2024.Effects of 2D/3D urban morphology on land surface temperature: Contribution, response, and interaction.*Urban Climate*,**53**,101791. https://doi.org/10.1016/j.uclim.2023.101791 (2024).
56. Gupta K. Seasonal analysis of the relationship between 2D/3D Urban Canopy Parameters, Land Surface Temperature and Ventilation Potential[C]//AGU Fall Meeting Abstracts. 2023, 2023: GC43B-06.
57. Liu Y, Wang Z, Liu X, et al. Complexity of the relationship between 2D/3D urban morphology and the land surface temperature: a multiscale perspective. *Environmental Science and Pollution Research*,**28**,66804–66818. https://doi.org/10.1007/s11356-021-15177-7(2021).

Acknowledgements
Acknowledgement for the data support from "National Tibetan Plateau / Third Pole Environment Data Center (http://data.tpdc.ac.cn)".

Author contributions
Caiyi Huang wrote the main manuscript text,implemented the code,analyzed the result,and created charts; Ke Liu provided writing-review and editing,and supervision; Tianyi Ma provided writing-revise; Hang Xue implemented and verified the code; Pengfei Wang and Lin Li obtained and verified the data; All authors reviewed the manuscript.

Funding
Funding for this work was provided by the National Natural Science Foundation Program (No. 42201244).

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to K.L.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.